UNIT 4 · 8 hours

Root Locus: Concept and Conditions



Stability of Systems

Control Systems (22EEG53)
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NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

- Define the root locus
- State the magnitude and angle conditions
- Relate poles and zeros of G(s)H(s) to the locus
- Interpret the locus for a simple system

Syllabus context

Stability of systems: characteristic equation, location of roots, absolute and relative stability, Routh–Hurwitz criterion (necessary and sufficient conditions), root locus concept, rules for construction of root locus.

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What is the root locus?

Definition: the plot of closed-loop poles in the s-plane as a parameter K varies from 0 to ∞.

Proposed by W. R. Evans (1948).

Why? shows how the transient response (pole positions) changes with gain K, without solving the characteristic equation repeatedly.

Starts from open-loop poles (K = 0) and ends at open-loop zeros or infinity (K → ∞).

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Conditions on the root locus

$$1 + K\,G1(s) = 0$$

characteristic equation, G1 = Π(s + zi)/Π(s + pj)

$$\angle G1(s)H1(s) = \pm 180°(2q + 1)$$

angle condition

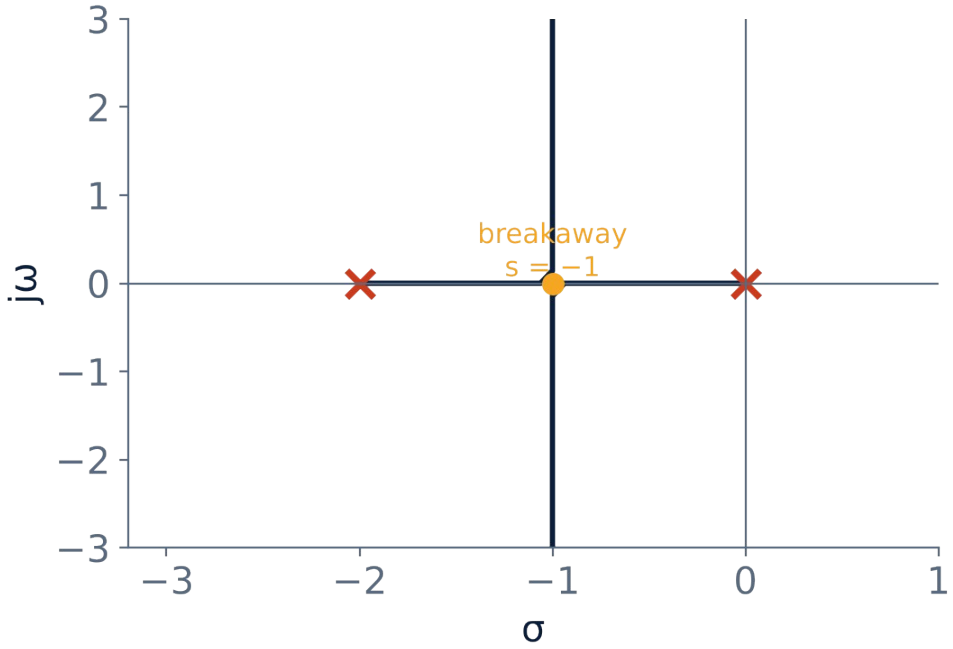
$$|K\,G1(s)H1(s)| = 1$$

magnitude condition (gives K)

- Angle condition decides whether a point lies on the locus.
- Magnitude condition gives the gain K at that point: K = Π|s + pj| / Π|s + zi|.

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Example: G(s)H(s) = K/[s(s + 2)]



- Poles at 0 and −2 (K = 0).
- Characteristic: $s^2 + 2s + K = 0$, $s = -1 \pm \sqrt{(1 - K)}$.
- 0 < K < 1: real roots, overdamped.
- K = 1: breakaway at s = −1.
- K > 1: complex roots on Re s = −1; ζ falls as K rises.

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Numerical: finding K on the locus

Problem For the locus above find K for $\zeta = 0.5$.

- $s^2 + 2s + K = 0 \Rightarrow \omega n = \sqrt{K},\ 2\zeta\omega n = 2 \Rightarrow \omega n = 2$
- $K = \omega n^2 = 4$
- Roots: $s = -1 \pm j\sqrt{3}$ (on the 60° line)

Answer: K = 4, poles −1 ± j1.73

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Effect of adding poles and zeros

Adding a pole

Pushes the locus to the right: reduces stability and relative damping.

Adding a zero

Pulls the locus to the left: improves stability and damping.

Use in design

Lead compensator adds a zero (left), lag adds a pole-zero close to the origin.

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Key takeaways

- Root locus: closed-loop poles vs K (0 → ∞)
- Angle condition: ±180°(2q + 1); magnitude condition gives K
- Starts at open-loop poles, ends at zeros or infinity
- Adding poles destabilises; adding zeros stabilises

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Practice questions

Q1. State the angle and magnitude conditions.

Q2. Sketch the root locus of K/[s(s + 4)] and find K for critical damping.

Q3. What is the effect of adding a pole at −3 to K/[s(s+2)]?

Q4. Find the point on the real-axis locus where K = 1 for K/[s(s+2)].